Projection-Based Fusion of Semantic and Collaborative Embeddings for Long-Tail Recommendation

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Abstract. Long-tail items in recommender systems often receive limited exposure as collaborative models are biased towards popular items having dense historical interaction patterns. Content-based representations can enrich collaborative representations with additional semantic information; however, their effective integration requires careful alignment of the distinct information captured by the two representation spaces. To address the long-tail problem in recommender systems, we propose a semantic–collaborative item representation scheme that integrates BERT-based semantic features from reviews and textual attributes with LightGCN-based collaborative signals from user–item interactions. This integration provides richer representations for sparsely interacted tail items and enhances their recommendation potential. We evaluate two fusion strategies: fixed-weight fusion of the 64-dimensional BERT and LightGCN embeddings, and projection-based fusion, where a learnable bias-free linear transformation is applied to the BERT embedding before combining it with the LightGCN embedding. Experiments are conducted on the real-world dataset MovieLens-1M together with textual movie metadata. Experimental results show the effectiveness of the proposed approach over the benchmark models in terms of NDCG and Recall.

Keywords: Recommender systems · Long-tail recommendation · LightGCN · BERT · Hybrid fusion.

1 Introduction

Recommender systems help users discover relevant items from large collections such as movies, books, music, and e-commerce products [20]. The two most widely adopted approaches are Collaborative Filtering (CF) and Content-Based (CB) filtering [9, 17]. CF infers latent preferences from past user–item interactions; matrix factorization [15] and graph-based models such as LightGCN [8] perform strongly, but they depend on historical interactions and suffer from data sparsity and popularity bias, where frequently interacted items dominate recommendation lists while long-tail items receive limited exposure [1]. CB filtering instead uses descriptive attributes such as plot summaries, genres, cast information, and user reviews [17], which remain available when interactions are

sparse, but it does not exploit collaborative preferences and its accuracy can be limited [17, 4].

Hybrid systems combine both signals [4]. This paper proposes a hybrid movie recommendation framework that combines collaborative item representations learned by LightGCN [8] with semantic item representations produced by a frozen pretrained BERT model [5] from movie plots, genres, cast and crew information, and user reviews. Two fusion mechanisms are studied: a fixed-weight *Static Hybrid* and a *Projection Hybrid* that learns a bias-free linear alignment of the semantic representation into the collaborative embedding space. The framework is evaluated on MovieLens-1M [7] enriched with textual metadata, under a per-user chronological train–validation–test split and an implicit-feedback Top-K protocol, against BPR-MF, LightGCN and the self-supervised SGL-ED baseline [24].

The contributions are: (i) a hybrid framework integrating LightGCN and BERT; (ii) rich semantic item representations built from plots, genres, cast, crew and reviews; (iii) two fusion strategies, static and projection-based, and an analysis of their effect on ranking quality and catalog exposure; (iv) experiments showing that semantic–collaborative fusion substantially improves long-tail retrieval relative to LightGCN while revealing a trade-off between ranking effectiveness and catalog exposure.

2 Related Work

Collaborative and graph-based recommendation. Neighborhood methods exploit user or item similarities [9, 21], matrix factorization learns latent factors from observed interactions [15], with extensions for regularization and temporal dynamics [14]. Such models amplify popularity bias, since frequently interacted items provide stronger training signals [1]. Graph-based models propagate information along observed user–item edges [25]: NGCF introduced message passing for higher-order signals [23], and LightGCN removed feature transformations and nonlinear activations [8]. They remain dependent on interaction density, so rarely interacted items are hard to retrieve [8, 1].

Content-aware recommendation. CB methods exploit genres, keywords, descriptions and reviews [17]; review text encodes latent dimensions relevant to recommendation [16], and pretrained transformers such as BERT provide contextualized text representations [5]. Content alone does not encode the collaborative preference structure [17, 4].

Hybrid and long-tail recommendation. Hybrid systems combine both signals [4], which matters for the long tail because semantic information remains available when interaction evidence is weak [17, 16]. SGL adds self-supervised learning over perturbed graph views with a contrastive objective and explicitly considers the long-tail limitations of graph recommendation [24], improving the collaborative learning process itself. In contrast, this work asks whether aligning a fixed semantic item representation with the LightGCN embedding space im-

proves long-tail retrieval, comparing fixed weighting against a learned projection under one protocol.

3 Preliminaries

3.1 LightGCN Collaborative Representation

Training interactions form a bipartite graph $\mathcal{G} = (\mathcal{U}, \mathcal{I}, \mathcal{E})$ with an edge $(u, i) \in \mathcal{E}$ for every observed user–item pair. With $\mathbf{R}$ the user–item interaction matrix, LightGCN [8] uses the adjacency matrix and its symmetric normalization

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{0} \end{bmatrix}, \qquad \hat{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}}, \tag{1}$$

where $\mathbf{D}$ is the diagonal degree matrix of $\mathbf{A}$. Each user and item has a trainable embedding [8], stacked as $\mathbf{E}^{(0)} \in \mathbb{R}^{(|\mathcal{U}|+|\mathcal{I}|)\times d}$ with $d = 64$. Propagation and the final representation are

$$\mathbf{E}^{(k+1)} = \hat{\mathbf{A}} \mathbf{E}^{(k)}, \qquad \mathbf{E} = \frac{1}{K+1} \sum_{k=0}^{K} \mathbf{E}^{(k)}, \tag{2}$$

with $K = 3$ layers and no feature transformation or nonlinear activation. The collaborative score of user u for item i is $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$. The model is trained with Bayesian Personalized Ranking (BPR) [18]: for a triple (u, i, j) with observed positive item i and sampled negative item j,

$$\mathcal{L} = -\frac{1}{|\mathcal{B}|} \sum_{(u,i,j)\in\mathcal{B}} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda \left(\left\| \mathbf{E}_U^{(0)} \right\|_2^2 + \left\| \mathbf{E}_I^{(0)} \right\|_2^2 \right), \tag{3}$$

where $\mathcal{B}$ is a mini-batch, $\sigma(\cdot)$ the sigmoid function and $\lambda = 10^{-5}$ the L_2 coefficient [8]. Negatives are sampled uniformly from the catalog excluding the user's training items [18], five per positive. Training uses Adam [13] with learning rate 10^{-3}, mini-batch size 4096, at most 200 epochs and random seed 42. The resulting $\mathbf{E}_U$ (6040×64) and $\mathbf{E}_I$ (3706×64) are kept fixed for the hybrid models.

3.2 Content-Aware Item Representation

For every movie a document is built by concatenating its title, genre labels, plot summary, tagline, cast descriptions, crew information and user reviews; missing fields are replaced with empty strings. The document is tokenized and split into chunks of at most 510 tokens. Each chunk receives the `[CLS]` and `[SEP]` tokens, is passed through a frozen pretrained BERT-base encoder, and the final hidden state of `[CLS]` is extracted [5]; chunk vectors are mean-pooled into $\mathbf{h}_i \in \mathbb{R}^{768}$. Because the BERT and LightGCN dimensionalities differ, a fixed random projection [3] $W_B \in \mathbb{R}^{768\times 64}$, sampled from a zero-mean Gaussian with standard deviation 0.01 under seed 42, gives $\tilde{\mathbf{e}}_i^{\mathrm{BERT}} = W_B^\top \mathbf{h}_i \in \mathbb{R}^{64}$. This step reduces dimensionality only and is not the learned alignment of the Projection Hybrid; BERT is not fine-tuned, and the 64-dimensional vectors are kept fixed for both hybrid variants.

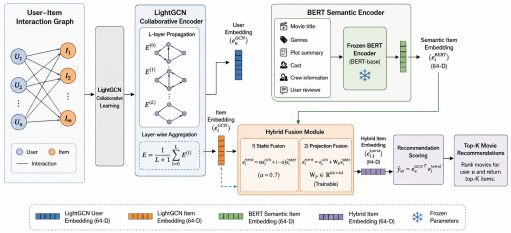


Fig. 1: Overview of the framework: LightGCN embeddings from the interaction graph and frozen-BERT semantic item representations from movie text are combined by static or projection-based fusion for top-K recommendation.

4 Proposed Method

Figure 1 shows the framework, which consists of three stages. The *collaborative stage* trains LightGCN on the training graph and stores $\mathbf{E}_U$ and $\mathbf{E}_I$. The *semantic stage* encodes each movie document with the frozen BERT encoder and reduces the output to a fixed 64-dimensional vector (Section 3.2). The *fusion stage* combines the two item representations into one hybrid item embedding that is scored against the unchanged LightGCN user embedding. The collaborative and semantic components are trained once and kept frozen; only the fusion stage differs between the two variants, and the only trainable component introduced is a single 64×64 matrix.

4.1 Problem Formulation

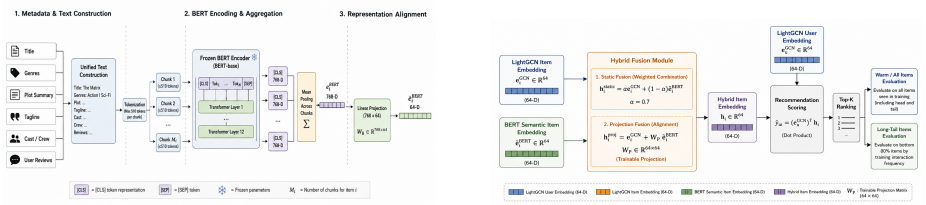
Let $U = \{u_1, \ldots, u_M\}$ be the users, $I = \{i_1, \ldots, i_N\}$ the movies and $E \subseteq U \times I$ the observed interactions, treated as implicit feedback: every observed pair is a positive signal. Let n_i be the number of training interactions of item i; ranking items by n_i in ascending order, the long-tail set $I_{\text{tail}} \subset I$ is the bottom 80% of the training catalog. The task is a Top-K ranking over the full catalog I in which relevant held-out items in I_{tail} are retrieved and ranked highly (Recall@K and NDCG@K on long-tail ground truth). Every model scores an item as $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$ with its own final item embedding $\mathbf{e}_i$. Let $\mathbf{e}_i^{\text{GCN}} \in \mathbb{R}^{64}$ denote the fixed LightGCN item representation and $\widetilde{\mathbf{e}}_i^{\text{BERT}} \in \mathbb{R}^{64}$ the fixed semantic representation.

4.2 Static Hybrid Fusion

Figure 2 shows the semantic pipeline and both fusion paths. The Static Hybrid model blends the two fixed representations by weighted linear interpolation,

$$\mathbf{e}_i^{\text{Static}} = \alpha\, \mathbf{e}_i^{\text{GCN}} + (1 - \alpha)\, \widetilde{\mathbf{e}}_i^{\text{BERT}}, \tag{4}$$

where $\alpha \in [0, 1]$ controls the relative contribution of collaborative and semantic information. The coefficient was selected in preliminary experiments over candidate values in $[0, 1]$; $\alpha = 0.7$ gave the best observed performance and was kept fixed for all reported results. Setting $\alpha = 1$ recovers the LightGCN item embedding exactly and $\alpha = 0$ ranks items purely by their semantic representation. The static model has no trainable parameters.



(a) (b)

Fig. 2: (a) BERT semantic pipeline: movie text is chunked (at most 510 tokens), encoded by frozen BERT-base, `[CLS]` vectors are mean-pooled and projected to 64 dimensions. (b) Item-embedding fusion: static weighting with $\alpha = 0.7$, or a trainable bias-free 64×64 projection followed by additive fusion.

4.3 Projection-Based Hybrid Fusion

The Projection Hybrid learns a transformation of the fixed semantic representation before combining it with the collaborative one. Since the semantic vector is already 64-dimensional, the trainable projection is $W_P \in \mathbb{R}^{64\times 64}$ with no bias term, and the item representation is obtained by additive fusion,

$$\mathbf{e}_i^{\text{Projection}} = \mathbf{e}_i^{\text{GCN}} + W_P\, \widetilde{\mathbf{e}}_i^{\text{BERT}}. \tag{5}$$

The LightGCN user and item embeddings and the BERT-derived embeddings remain frozen; only W_P, that is $64 \times 64 = 4{,}096$ parameters, is optimized with a pairwise BPR objective over triples (u, i, j),

$$\mathcal{L}_{\text{Projection}} = -\frac{1}{|\mathcal{B}|} \sum_{(u,i,j)\in\mathcal{B}} \log \sigma\Big(\mathbf{e}_u^\top \mathbf{e}_i^{\text{Projection}} - \mathbf{e}_u^\top \mathbf{e}_j^{\text{Projection}}\Big), \tag{6}$$

using Adam [13] with learning rate 10^{-3} for 10 epochs of 1,000 optimization steps each; negatives are sampled from the catalog excluding the user's training items. Training iterates over mini-batches of such triples and updates only W_P; after the last epoch the hybrid item matrix $\mathbf{E}_I + \widetilde{\mathbf{E}}^{\text{BERT}} W_P^\top$ is computed once and stored. At recommendation time, for every model, the items a user interacted with in training are removed from the candidate set [8], the remaining catalog items are scored by the inner product with the user embedding, and the top K are returned for $K \in \{10, 20, 50, 100\}$.

5 Experimental Setup

5.1 Dataset and Preprocessing

MovieLens-1M [7] contains 6,040 users, 3,706 movies and 1,000,209 user–movie ratings with timestamps, titles and genre labels. Plot summaries, taglines, cast, crew and available user reviews were collected from the TMDB API [22] and consolidated into one document per movie (Section 3.2). The 1–5 ratings were converted into implicit feedback, each observed interaction being a positive signal [11, 18]. Interactions were ordered chronologically per user and split into

Table 1: Training configuration of the compared models ("–": not applicable).

Parameter	LightGCN	BPR-MF	Static Hybrid	Projection Hybrid	SGL-ED
Embedding dimension	64	64	64	64	64
Graph layers	3	–	–	–	3
Learning rate	10^{-3}	10^{-3}	–	10^{-3}	10^{-3}
Batch size	4096	1024	–	–	2048
Epochs	200 (max)	10	–	10	1000
Optimization steps / epoch	–	1,000	–	1,000	–
L_2 regularization	10^{-5}	–	–	–	10^{-4}
Negative samples per positive	5	1	–	1	–
Fusion coefficient α	–	–	0.7	–	–
Projection W_P (bias)	–	–	–	64×64 (none)	–
SSL loss weight / temperature	–	–	–	–	0.1 / 0.2
Edge-drop ratio	–	–	–	–	0.1

715,442 training, 82,316 validation and 202,451 test interactions (all 1,000,209 accounted for). For warm-start evaluation, the 39 test interactions whose item is absent from training were excluded, leaving 202,412 evaluated test interactions. The long-tail set is defined from training-set frequencies only: the bottom 80% of the training catalog [2, 1]. Experiments ran on an Intel Core i5-12450H, 16 GB RAM and an NVIDIA RTX 3050 (4 GB) under Windows 11, seed 42.

5.2 Compared Models and Settings

All models use the same partitions, candidate masking and ranking procedure. Table 1 lists every training configuration; all trained models use Adam [13]. **BPR-MF** [18] is a matrix factorization baseline trained with the BPR loss and scored by dot product without L_2 normalization. **LightGCN** [8] is the primary collaborative baseline, **Static Hybrid** ($\alpha = 0.7$) an internal fixed-weight baseline, and **Projection Hybrid** the proposed model. **SGL-ED** [24] is an external self-supervised graph baseline retrained on the same split and evaluated with the same candidate masking, full-catalog ranking and Top-K protocol.

5.3 Evaluation Measures

Models are evaluated in an implicit-feedback, warm-start Top-K setting [19] with Recall@K, NDCG@K [12] and Catalog Coverage@K [10, 6], the fraction of the catalog appearing at least once across users' lists, for $K \in \{10, 20, 50, 100\}$. A user's complete held-out test set is the relevant set. The long-tail evaluation restricts the ground truth to items in I_{tail} while the candidate set remains the complete 3,706-item catalog, so the Coverage of LightGCN, BPR-MF, Static Hybrid and Projection Hybrid keeps the full-catalog denominator and measures overall exposure during long-tail evaluation. SGL-ED computes long-tail Coverage over the 2,964-item long-tail catalog, counting only recommended long-tail

Table 2: All-item performance on MovieLens-1M (full-catalog ground truth). Best value per column in bold.

	Recall				NDCG				Catalog Coverage			
Model	@10	@20	@50	@100	@10	@20	@50	@100	@10	@20	@50	@100
LightGCN [8]	**0.0681**	**0.1186**	**0.2416**	**0.3838**	**0.1648**	**0.1692**	**0.2038**	**0.2556**	0.2207	0.2963	0.4374	0.5672
BPR-MF	0.0511	0.0931	0.1925	0.3146	0.1134	0.1202	0.1501	0.1944	0.3559	0.4368	0.5685	0.6805
Static Hybrid ($\alpha = 0.7$)	0.0418	0.0768	0.1625	0.2732	0.0893	0.0983	0.1273	0.1687	**0.6937**	**0.7534**	0.8044	0.8419
Projection Hybrid (Ours)	0.0457	0.0846	0.1852	0.3148	0.1028	0.1121	0.1463	0.1954	0.3710	0.4684	0.6185	0.7299
SGL-ED [24]	0.0425	0.0713	0.1368	0.2123	0.0913	0.0951	0.1144	0.1424	0.5939	0.7458	**0.9228**	**0.9792**

Table 3: Long-tail performance on MovieLens-1M: Recall and NDCG use long-tail test interactions (bottom 80% of items by training frequency [1]) as ground truth; Coverage uses the full 3,706-item catalog, except SGL-ED (2,964-item long-tail catalog, not comparable). Best comparable relevance value in bold.

	Recall				NDCG				Catalog Coverage			
Model	@10	@20	@50	@100	@10	@20	@50	@100	@10	@20	@50	@100
LightGCN [8]	0.0016	0.0044	0.0191	0.0647	0.0027	0.0044	0.0108	0.0264	0.2207	0.2963	0.4374	0.5672
BPR-MF	0.0051	0.0144	0.0453	0.1007	0.0043	0.0082	0.0185	0.0346	0.3545	0.4352	0.5679	0.6794
Static Hybrid ($\alpha = 0.7$)	0.0258	0.0459	0.1044	0.1889	0.0347	0.0410	0.0598	0.0863	0.6937	0.7534	0.8044	0.8419
Projection Hybrid (Ours)	**0.0565**	**0.0965**	**0.1902**	**0.3018**	**0.0697**	**0.0811**	**0.1097**	**0.1437**	0.3710	0.4684	0.6185	0.7299
SGL-ED [24]	0.0170	0.0361	0.0837	0.1493	0.0138	0.0213	0.0373	0.0570	0.4673	0.6498	0.8877	0.9700

items, so its Coverage is reported for completeness but is not directly comparable; its Recall and NDCG are.

6 Results

6.1 All-item Performance

Table 2 and Figure 3(a) report performance over the complete catalog. LightGCN achieves the highest Recall and NDCG at every list size, with Recall@20 of 0.1186 and NDCG@20 of 0.1692. BPR-MF reaches Recall@20 of 0.0931 and NDCG@20 of 0.1202 with Coverage@20 of 0.4368. Static Hybrid raises Coverage@20 to 0.7534 but lowers Recall and NDCG. Projection Hybrid improves on Static Hybrid, reaching Recall@20 of 0.0846 and NDCG@20 of 0.1121 against 0.0768 and 0.0983, with Coverage@20 of 0.4684. SGL-ED obtains Recall@20 of 0.0713 and NDCG@20 of 0.0951 with Coverage@20 of 0.7458, rising to 0.9792 at $K = 100$.

6.2 Long-tail Performance

Table 3 and Figure 3(b) restrict the ground truth to long-tail items, and the picture changes substantially. LightGCN reaches only 0.0044 Recall@20 and 0.0044 NDCG@20, and BPR-MF 0.0144 and 0.0082: low-frequency items are hard to retrieve from collaborative signals alone. Static Hybrid raises these to 0.0459 and 0.0410. Projection Hybrid provides the strongest long-tail ranking among all evaluated models at every list size, with Recall@20 of 0.0965 and NDCG@20 of

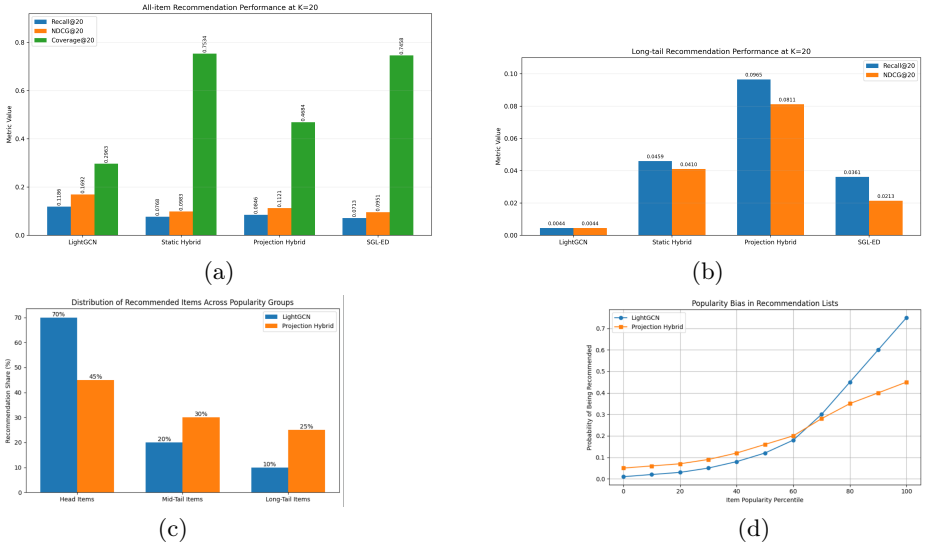


Fig. 3: Results at $K = 20$: (a) all-item Recall, NDCG and Coverage; (b) long-tail Recall and NDCG, highest for Projection Hybrid; (c) share of recommendations across three popularity groups; (d) probability of being recommended across popularity percentiles.

0.0811, approximately 21.9 times the Recall@20 of LightGCN; its Coverage@20 of 0.4684 equals its all-item value because the ground-truth restriction leaves candidates and ranking unchanged, so it measures full-catalog exposure, not the fraction of long-tail items covered. SGL-ED [24] reaches 0.0361 Recall@20 and 0.0213 NDCG@20 under the same protocol; Projection Hybrid retrieves a larger fraction of held-out long-tail interactions and ranks them higher at every reported K. SGL-ED's higher long-tail Coverage (0.6498 at $K = 20$) uses the 2,964-item long-tail denominator and is not directly comparable (Section 5.3).

6.3 Popularity Bias and Exposure

Coverage does not show how exposure is distributed over popularity, so Figure 3(c,d) analyses it directly. Panel (c) groups items into three popularity groups by training-set frequency (for visualization only; distinct from the bottom-80% definition of Table 3). At $K = 20$ LightGCN concentrates its recommendations on the most frequently interacted group, approximately 70/20/10 across the three groups, whereas Projection Hybrid allocates approximately 45/30/25. Panel (d) plots the probability of being recommended across popularity percentiles: LightGCN rises steeply toward the most popular items, while Projection Hybrid is flatter across the spectrum. Semantic alignment thus changes the popularity distribution of the lists, not only the long-tail metrics.

7 Conclusion and Future Work

This study combined LightGCN collaborative representations with BERT-derived semantic representations for long-tail recommendation through two fusion strategies, Static Hybrid and Projection Hybrid. On the complete catalog LightGCN ranks best, but its performance collapses on long-tail items. Projection Hybrid, which learns an alignment between the semantic and collaborative spaces, improves long-tail Recall@20 from 0.0044 to 0.0965 and NDCG@20 from 0.0044 to 0.0811, and outperforms the SGL-ED baseline on long-tail ranking. Static Hybrid gives the broadest coverage; Projection Hybrid keeps broader exposure than LightGCN with stronger ranking. Within the evaluated MovieLens-1M setting, semantic–collaborative alignment is a promising way to improve long-tail retrieval and reduce concentration on popular items.

The experiments use one dataset and a fixed seed; future work should cover more datasets and seeds, ablate the metadata fields, fusion weight, projection dimension and long-tail definition, compare the random projection with PCA or a trainable one, and add popularity-based and BERT-only baselines.

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